Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Superstore Sales Performance Analysis Dashboard

Submitted By:

Name: Abhinav

Abhilash

Reg. No.: 2411021240029

Date: 31/08/2026

GitHub Link: https://github.com/abhi7abhi/BA

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 1

Power BI Sales Performance Analysis – Sample Superstore

Problem Statement

Management wants to quickly understand sales performance across products, categories, sub-categories, and regions. The objective is to prepare the Sample Superstore dataset in Power BI and build an interactive dashboard that makes the major business patterns easy to identify.

Dataset and Model Used

- The PBIX contains a single main table named Sample_ Superstore.
- The table includes Order Date, Ship Date, Ship Mode, Customer ID, Segment, Country, City, State, Region, Product ID, Category, Sub-Category, Product Name, Profit, Quantity and Sales.
- The PBIX includes the DAX measures Total Sales, Total Profit and Total Quantity.
- Analysis is performed directly from the Sample_ Superstore table.

Step 1: Data Preparation and Import

- Opened the Sample Superstore data in Power BI Desktop.
- Inspected the imported table in Data view.
- Verified the required numeric, categorical, product, region and date fields.
- Prepared the dataset before developing the report visuals.

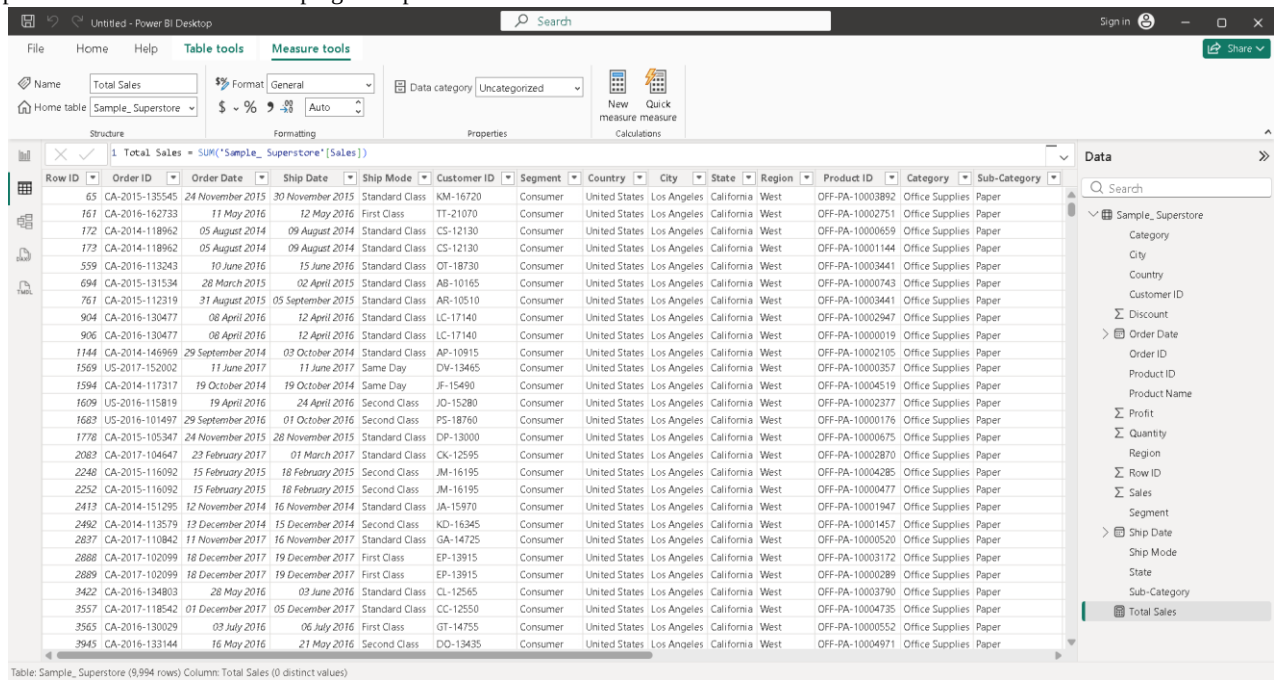


Figure 1: Sample_ Superstore table and Total Sales measure

Step 2: Data Cleaning and Transformation

- Opened Power Query Editor to inspect the Sample_ Superstore query.
- Checked the initial column types after headers were promoted.
- Converted date fields to appropriate Date types for analysis.
- The Ship Date transformation uses the English (United States) locale.
- Reviewed the transformed data preview before loading it into the report.

The Power Query screenshots show the preparation stage. Order Date and Ship Date are treated as dates, while Order ID, Ship Mode, Customer ID and Segment remain categorical/text fields.

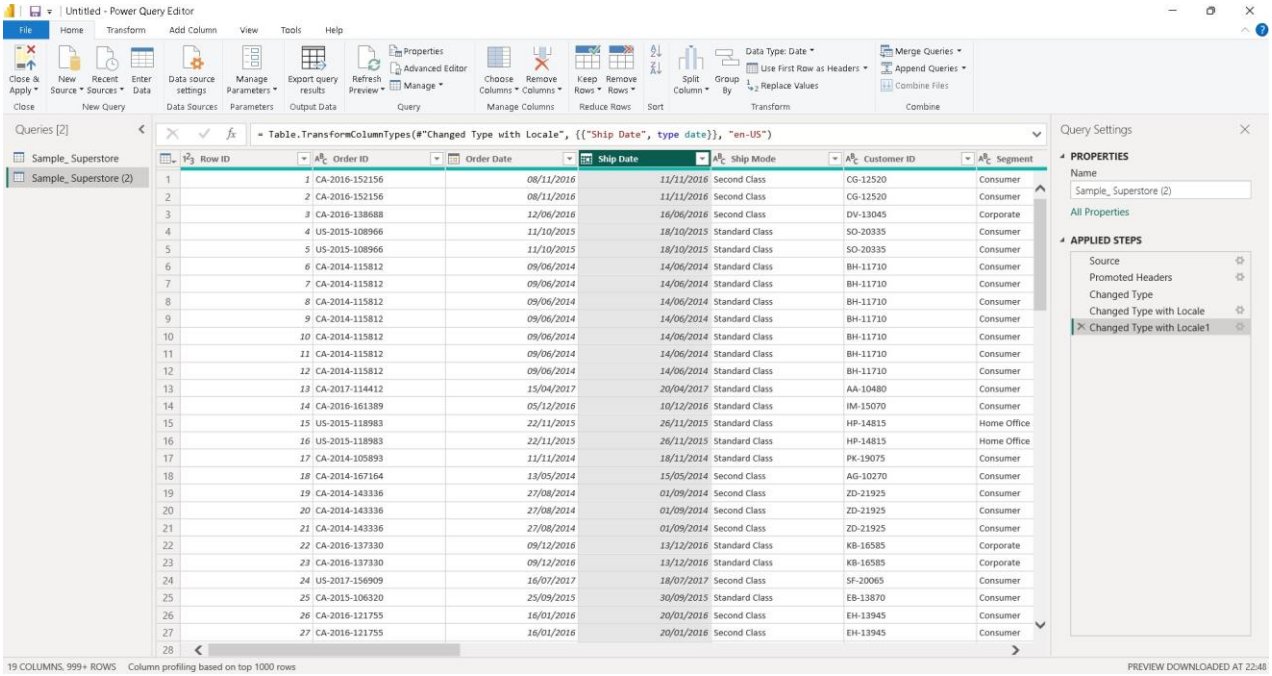


Figure 2: Power Query – Ship Date date-type transformation

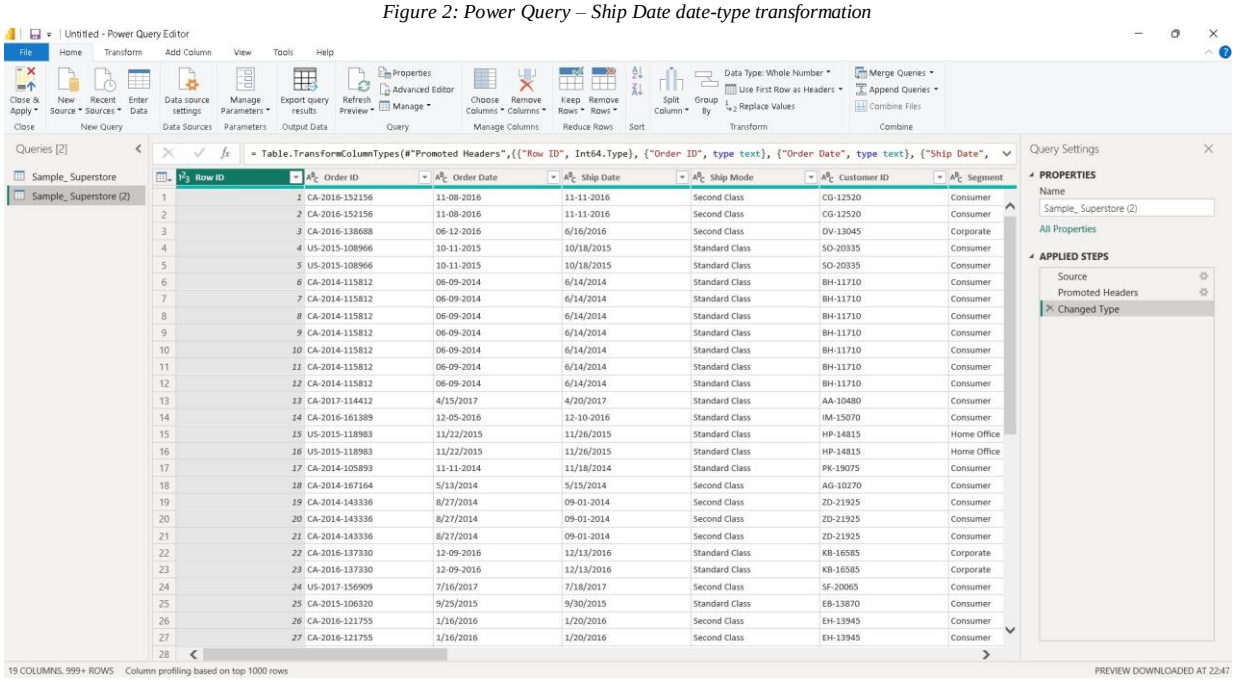


Figure 3: Power Query – promoted headers and changed types

Step 3: DAX Measures

- Total Sales – sum of Sales.
- Total Profit – sum of Profit.
- Total Quantity – sum of Quantity.

Step 4: Power BI Visualizations

The dashboard was developed to answer management questions about overall performance, category performance, regional performance, products and time trends.

- Total Sales by Category – horizontal bar chart.
- Total Sales by Region – column chart.
- Total Sales by Sub-Category – 100% stacked bar chart.
- Total Sales by Product Name – bar chart.
- Total Sales and Total Profit – scatter chart.
- Total Sales by Year – line chart.
- Region, Year and Category – interactive slicers.
- KPI cards – Total Sales, Total Profit and Total Quantity.

Visualization Pictures

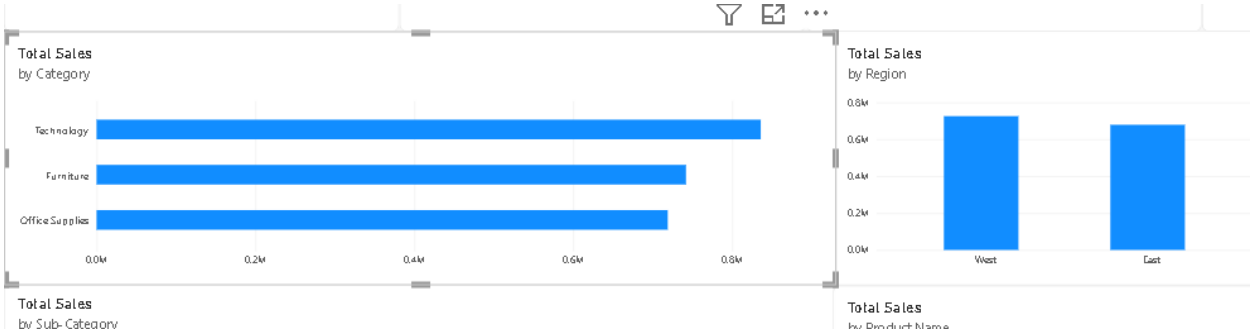


Figure 4: Sales by Category and Sales by Region

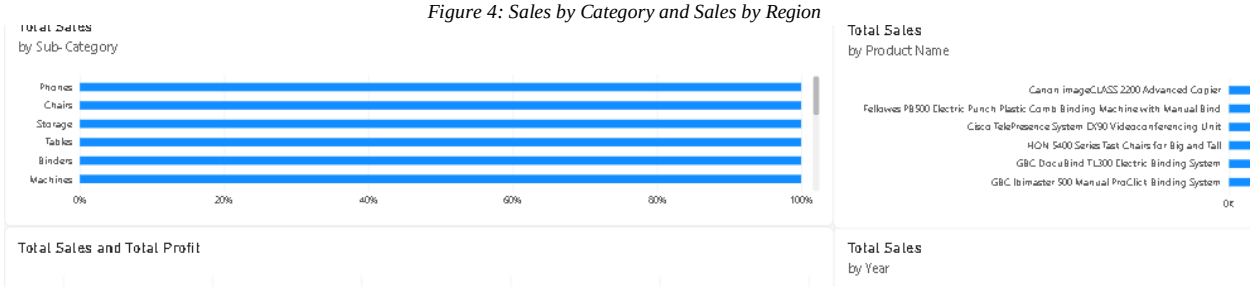


Figure 5: Sales by Sub-Category and Product Name

Step 5: Dashboard Development and Output

The final PBIX page combines KPI cards, comparative charts and interactive slicers into one management dashboard.

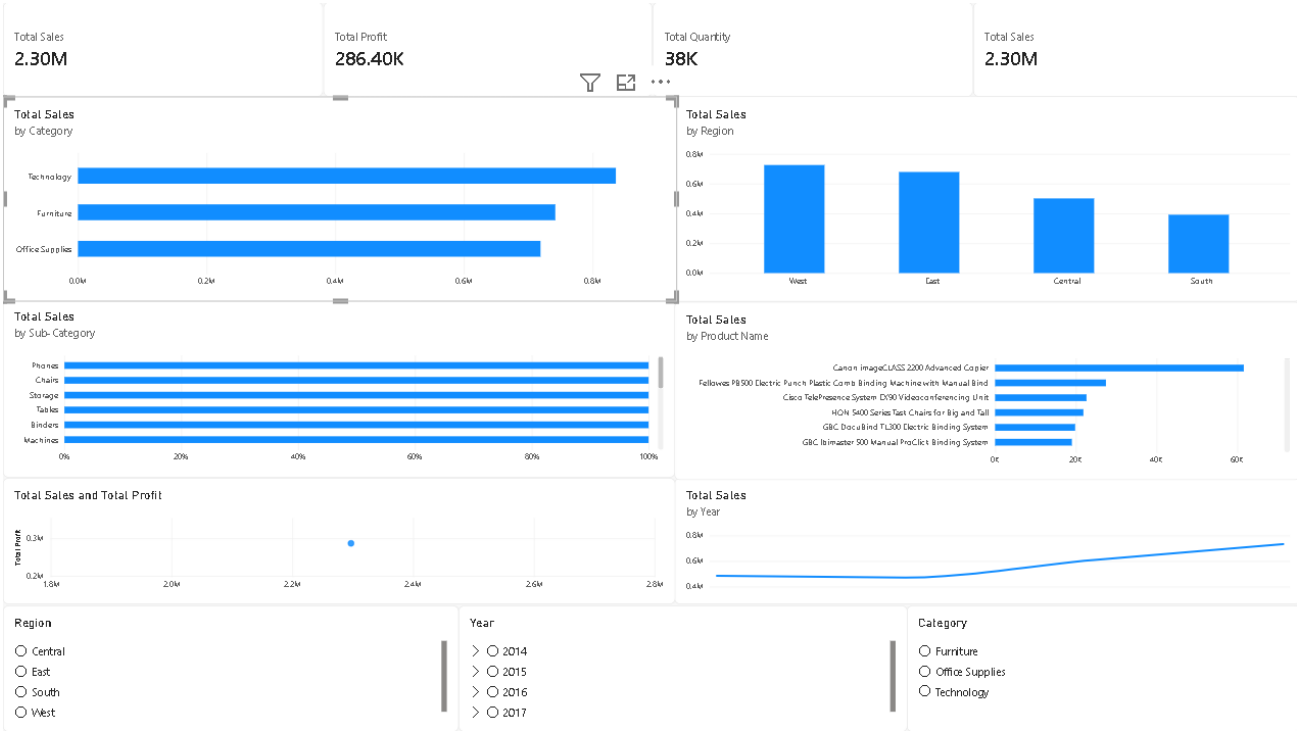


Figure 6: Complete Sample Superstore Power BI dashboard

Dashboard KPI Values

- Total Sales: 2.30M
- Total Profit: 286.40K
- Total Quantity: 38K

Trend and Interactive Filtering

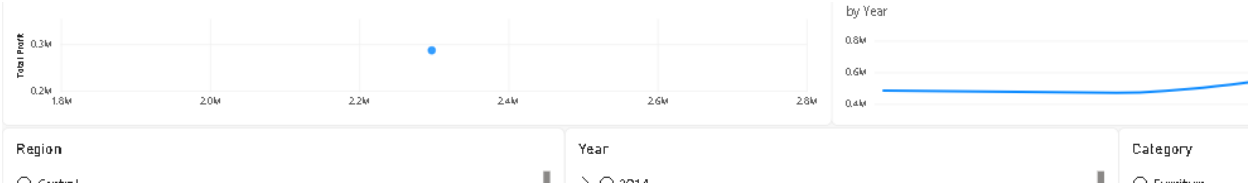


Figure 7: KPI cards, Sales vs Profit and yearly sales trend



Figure 8: Region, Year and Category slicers

Business Insights, Conclusion and Learning Outcomes

Observed Patterns

- Technology is the highest-selling category in the displayed category chart, followed by Furniture and Office Supplies.
- West has the highest regional sales, followed by East, Central and South.
- The Product Name chart identifies leading products by Total Sales.
- The yearly line chart shows changes in Total Sales over time.
- The scatter visual compares Total Sales with Total Profit.

Business Use

- Management can compare category performance quickly.
- Regional performance can be evaluated using the Region visual and slicer.
- Product-level sales can be used to identify leading products.
- Year and Category slicers allow focused analysis.

Conclusion

This project successfully uses the Sample Superstore dataset to create an interactive sales-performance report in Power BI. Data preparation was performed in Power Query, DAX measures were used for key metrics, and multiple visualization types were selected to answer business questions. The final dashboard combines sales, profit, quantity, category, sub-category, product, regional and yearly analysis.

Learning Outcomes

- Learned Power BI data inspection and preparation.
- Learned Power Query data-type and transformation tasks.
- Learned to create and use DAX measures.
- Learned to create KPI cards, charts and slicers.
- Learned to interpret sales patterns across categories, products, regions and years.

Final Result

The completed PBIX report provides management with a centralized and interactive view of Sample Superstore sales performance and supports quick comparison across products, categories, sub-categories, regions and years.